Paper code: CSC-503/ CEN-505 (Old) Roll No. 23BCS066

B. Tech. (Computer Engineering) V Semester Examination, 2025-26

Time: 3 Hours **Computer Networks (Paper code: CSC-503/ CEN-505 (Old))** **Max. Marks: 45/60**

Note: Attempt all questions. Make suitable assumption(s) wherever required.

Q1. (a) **CO1** Compare and discuss how each of the following layers — Physical, Data Link, Network, and Transport, helps in transferring a file from one laptop to another across different LANs connected by routers. **[4.5/6] [BL1,3]**

Explain:

(i) The data unit used at each layer
(ii) The address type used for identification
(iii) The networking devices involved
(iv) The type of data delivery performed by each layer

Draw a simple diagram showing how data moves across layers during file transfer.

(b)(i) In the TCP/IP protocol suite, explain how the transport layer provides reliable data delivery even when the underlying network layer (IP) is unreliable and connectionless. **[2+2.5/ 3+3] [BL2,4]**

(ii) Comment on the following statements with proper justification:

(A) Packet switching leads to better utilization of bandwidth resources than circuit Switching.
(B) Packet switching results in less variation in delay than circuit switching.
(C) Packet switching requires more per packet processing than circuit switching.
(D) Packet switching can lead to reordering unlike in circuit switching.

OR

(b')(i) What problems could arise if IP addressing is correct but MAC addressing is incorrectly mapped due to an outdated ARP cache? How might this affect port-based communication? **[2.5+2 3+3 [BL2,4**

(ii) How does the absence of flow control at the Data Link layer lead to buffer overflows at the Transport layer? Discuss the cascading effects.

Q2. (a) **CO2** In the Go-Back-N ARQ protocol, explain why the sender's window size is 2n-1, while the receiver's window size is limited to 1. Additionally, describe how the protocol handles the following scenarios: (i) Duplicate frames (ii) Lost frames (iii) Lost acknowledgments **[4.5/([BL3,**

(b) The following character encoding is used in a data link protocol: **[4.5/([BL3,**
A:01000111; B:11100011; FLAG: 01111110; ESC: 111000000
Show the bit sequence transmitted (in binary) for the four-character frame: A ESC D FLAG when each of the following framing methods are used:
(i) Character Count
(ii) Flag bytes with byte stuffing
(iii) Starting and ending flag bytes, with bit stuffing

OR

(b') Consider a sliding window flow-control protocol operating between a sender and a receiver over a full-duplex, error-free communication link. The following assumptions hold:

- The processing time for the data frame at the receiver and for the acknowledgment frame at the sender is negligible.
- The sender always has an infinite number of frames ready for transmission.
- Each data frame has a size of 2000 bits, and each acknowledgment (ACK) frame has a size of 10 bits. The link data rate in both directions is 1 Mbps.
- The one-way propagation delay of the link is 100 milliseconds.

Determine the minimum sender's window size (in number of frames), required to achieve a link utilization of 50%.

Q3. CO3 (a) Describe in detail how the RTS/CTS (Request to Send / Clear to Send) mechanism in IEEE 802.11 MAC protocol helps in avoiding collisions caused by Hidden Terminals. Why can the RTS/CTS mechanism not fully eliminate the Exposed Terminal problem? [4.5/6] [BL1,2]

(b) In a TDM medium access control bus LAN, each station is assigned one time slot per cycle for transmission. Assume that the length of each time slot is the time to transmit 100 bits plus the end to end propagation delay. Assume a propagation speed of 2 * 10^ 8 m/sec. The length of the LAN is 1 km with a bandwidth of 10Mbps. Find the maximum number of stations that can be allowed in the LAN so that the throughput of each station can be 2/3 Mbps. [4.5/6] [BL4,5]

OR

(b') Consider a network using the pure ALOHA protocol. Each frame has a length of 1000 bits, and the channel transmission rate is 2 Mbps. The total number of transmission attempts (new + retransmissions) across all stations is modeled as a Poisson process with a rate of 600 frames per second. Find the throughput of the network in frames per second and in bits per second. [4.5/6] [BL5]

Q4. CO4 (a) Consider three machines M, N, and P with IP addresses 100.10.5.2, 100.10.5.5, and 100.10.5.6 respectively. The subnet mask is set to 255.255.255.252 for all the three machines. Which one of the following is true? Justify your answer. [4.5/6] [BL4

(A) M, N, and P belong to the same subnet.
(B) M and N belong to the same subnet; P belongs to a different subnet.
(C) M and P belong to the same subnet; N belongs to a different subnet.
(D) M, N, and P belong to different subnets.

(b) In a scenario where multiple devices are behind a NAT (Network Address Translation), explain how IP addresses and port numbers are translated. Show the scenario where packet is going out of a network address translation (NAT) device from an internal network to an external network? [4.5/([BL2,

OR

Give answer to the following questions based on IPv4 addresses: **[4.5/6]**

i) What is the purpose of the 127.0.0.1 address? Can it be used to communicate with another host? **[BL4]**

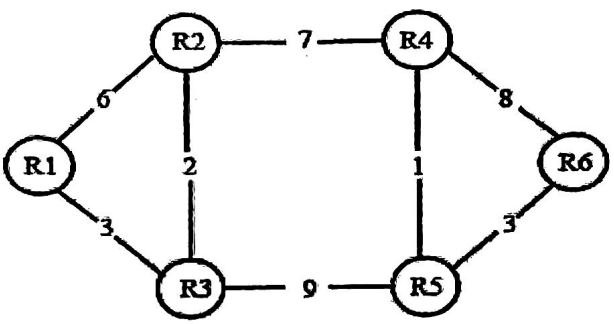
ii) A host sends a packet with destination 0.0.0.0. What happens?

iii) Differentiate between 255.255.255.255 and a subnet-directed broadcast address (for example, 192.168.1.255 in a /24 network).

iv) Explain the purpose of 224.0.0.0/4 range

v) Is 172.32.5.1, a special address or not?

Q5. (a) CO5 Consider a network with 6 routers R1 to R6 connected with links having weights as shown in the following diagram **[4.5/6] [BL4,5]**



All the router use the distance vector base routing algorithm to update their routing tables. Each router starts with its routing table initialized to contain an entry for each neighbor with the weight of the respective connecting link. After all the routing tables stabilize, how many links in the network will never be used for carrying any data?

(b) Explain in detail why OSPF (Open Shortest Path First) cannot be used for inter-domain routing, while BGP (Border Gateway Protocol) is specifically designed for that purpose. Discuss the fundamental differences in their routing policies, and the way each protocol handles routing information across networks belonging to different organizations. **[4.5/6] [BL2,4]**

OR

(b) Differentiate between static and dynamic routing. For the network given below, obtain the routing table of router B using link state routing. **[4.5/6 [BL2,6**

